Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/01/2014
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Molecular Biology of Growth and Cell Remodeling within the Spaceflight Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	robferl@ufl.edu	**Fax:**	FY 352-392-4072
PI Organization Type:	UNIVERSITY	**Phone:**	352-392-1928
Organization Name:	University of Florida		
PI Address 1:	Horticultural Sciences, 1301 Fifield Hall		
PI Address 2:	Box 110690		
PI Web Page:	http://www.hos.ufl.edu/		
City:	Gainesville	**State:**	FL
Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	08/01/2012	**End Date:**	12/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 12/31/2015 per NSSC information; original end date 7/31/13 (Ed., 6/24/14) NOTE: End date is now 6/30/2014 per NSSC information; original end date 7/31/13 (Ed., 8/9/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paul, Anna-Lisa (University of Florida)		
Grant/Contract No.:	NNX12AN69G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants experiencing spaceflight are quite normal in appearance but can exhibit growth habits distinctly different from plants on Earth. Historically the spaceflight-induced differences were difficult to dissect due to changing hardware and flight conditions. With the Advanced Biological Research System (ABRS) and a consistent ISS orbital environment, it is now possible to confidently approach a dissection of the molecular biology guiding the altered growth of plants in spaceflight. This proposal involves experiments that specifically address growth and molecular changes in Arabidopsis that occur during spaceflight, bringing the molecular and genetic tools of Arabidopsis to bear on fundamental questions of root morphology, growth, and cell wall remodeling. During the APEX-TAGES experiments, we observed a remarkable number of gene expression changes during spaceflight that are associated with cell wall restructuring, especially in roots. We also observed, with the ABRS/GIS imaging system, genetically dependent changes in root growth direction, movement, and structure. The current experiments will employ the ABRS and the GIS to dissect and better understand the molecular changes guiding growth restructuring on orbit. In particular we hypothesize that spaceflight conditions engender a constant state of root cell wall modification during the cell file rotations that characterize spaceflight-induced root skewing. The specific experiments will involve mutants in root growth and cell wall signaling pathways identified in our previous spaceflight molecular studies, as well as new Green Fluorescent Protein biosensor lines also based on the genes identified in the previous flight experiments. The overall goal of our program is to understand fundamental molecular biological responses to extraterrestrial environments. This goal aligns with Fundamental Space Biology goals to understand the impact of spaceflight on biological systems to better enable the exploration imperative. We have used molecular biology and genetically tagged plants as biological monitors of spaceflight and space-related environments with great science return, leading to the development of new biological and hardware tools to study the spaceflight response with even higher fidelity in a wider variety of spaceflight-related environments. The focus of the current proposal is to build on these data and other insights gained from the execution of the APEX-TAGES experiments in the ABRS /GIS, advancing science while also advancing the telemetric imaging hardware and biological experiment support for the ISS and future platforms Gaining insights into such mechanisms is recognized as fundamental within the decadal study and underpins answers to some of the biggest questions in spaceflight plant biology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To gain fundamental insight into how plants perceive and process the signals that control tropism and adaptive physiology in novel and extreme environments. It also highlights how the unique research environment of the ISS can provide can address biological questions that cannot be answered on Earth where gravity would mask many of the underlying phenomena.
Task Progress:	The major operations of the past year's reporting period was the Science Verification Test conducted in the ABRS / GIS hardware in the fall of 2013, and then the initial explorations into using alternative plant growth hardware (VEGGIE) in the early part of 2014. The Science Verification test (SVT) was conducted in two parts – one at Kennedy Space Center (KSC) in the ISSES orbital environment simulator chamber, and a parallel study in the PIs' laboratory at the University of Florida (UF-SVT). Three sets of evaluation were used: 1-general morphology of the plants (growth patterns, health and vigor), 2-RNA yield and quality, 3- expression of select genes. We used 12 day old plants as the primary developmental stage (and the only developmental stage in the KSC-SVT), but also included younger developmental ages in the UF-SVT ops (5 day and 8 day old plants). Plants were delivered (turned-over) as dormant, Duvetyne cloth-wrapped plates at ambient temperature. Ten plants were turned over: one imaging plate (position 1), five primary plates for positions 2-6, and four spares. Plates were launched 48 hours after turn over and inserted into GIS 5 days after Turnover. Plate 1 was images every 6 hours for duration of the experiment. The plates were harvested 12 days after insertion into GIS (one full plate to one KFT). The KFTs were allowed to set a room temperature for at least 12 hours, and transferred to -80 to mimic MELFI. The UF-SVT parallel study was designed solely for science and did not test on orbit operations. The VEGGIE compatibility test demonstrated that although a substantially different growth platform than ABRS, it could be adapted to the plant support requirements defined by this proposal. Lessons Learned from SVT. Better Ground Control images that match orbital in quality. A large amount of information is derived from the images taken of the plants just prior to harvest; growth patterns, morphology and morphometric data, as well as a gauge of general health and vigor of the plants. It is imperative that the harvest photos taken on orbit match the ones taken on the ground. Facile data delivery system The Nomad system used for SVT looks like it will work for data transfers. It is important to set the system up ahead of time, test accessibility of off-site PIs and ensure that it is updated regularly. Light levels . It is clear from the morphology of the plants at harvest that the plants were not receiving the same functional light levels as those plants run in APEX-01. It appears that the top tier plates of the SVT received close to the same amount of light as the bottom tier plates in APEX-01. The light level reaching the second tier plates was inadequate to support healthy growth. The APEX-01 plates in Figure 15 were 13.5 days old at harvest. The lighting problem needing resolution for PVT. Video support . Need better video support for local and ISS actions. What can we view at UF better than the slow updates? Insights into the developmental stages of Arabidopsis. Given the results of our SVT-related experiments at UF with respect to plant age and gene expression profiles, we have requested, and have been approved for, a change in the harvest scenarios. The new harvest schedule for the three runs will be after 5 days, 8 days, and 12 days respectively. The ERD will be revised to reflect this change, and the PVT will also accommodate the new harvest schedule. Conclusions from VEGGIE test. The standard Laboratory vertical plate form factor actually fits well within the VEG hardware. Three standard 10cm x 28cm growth racks fit within the base of VEGGIE; each rack has capacity for 10 10cm square plates. The habitat provided by VEGGIE is substantially different from that provided by ABRS; primary among the differences is the lighting quality and the internal environment. The internal environment freely exchanges with cabin air. Although the bellows walls provide some mitigation of relative humidity, gaseous mixtures (CO2, VOCs) cannot be controlled. Lighting is provided by a bank of red, blue and green LEDs located in the top of the unit. The intensity can be varied, but the quality and intensity cannot be tailored to match ABRS. However, initial measurements suggested that it well within the tolerances for arabidopsis. The VEGGIE compatibility test demonstrated that although a substantially different growth platform than ABRS, it could be adapted to the plant support requirements defined by this proposal.

Bibliography Type:	Description: (Last Updated: 11/27/2022)
Articles in Peer-reviewed Journals	Denison FC, Gökirmak T, Ferl RJ. "Phosphorylation-related modification at the dimer interface of 14-3-3 omega dramatically alters monomer interaction dynamics." Arch Biochem Biophys. 2014 Jan 1;541:1-12. Epub 2013 Nov 5. http://dx.doi.org/10.1016/j.abb.2013.10.025 ; PubMed PMID: 24211434 , Jan-2014